

Checkpoint 1: Dataset Setup

Objective

Verify the paired underwater dataset and define the supervised restoration task.

Key Message

The active protocol uses grouped source assignments and exact-pair deduplication for more defensible training and testing.

Completed Work

- Verified 3672 matched clear/turbid image pairs.
- Created contiguous source-group split files and a duplicate audit.
- Separated paired training data from unpaired qualitative data.

Verification and Controls

- Checked filename correspondence, readable image dimensions, and one-to-one source-target pairing before training.
- Removed 32 exact duplicate records before assigning whole content groups to a single split.
- Saved deterministic manifests so every later model uses the same 2,899/357/384 partition.

Evidence

Split	Pairs
Train	2899
Validation	357
Test	384

Decision / Next Gate

Freeze the grouped manifests; use validation for model selection and reserve the test set for final reporting.

Notes

- Quantitative metrics are valid only on paired clear/turbid data.

Turbid underwater patch



Clear reference patch



Verified paired sample: filename 100.jpg

Example verified paired image patch.

Pair 1500.jpg



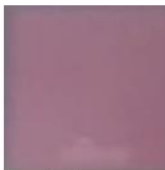
turbid

Pair 2750.jpg



turbid

Pair 2000.jpg



turbid

Pair 1750.jpg



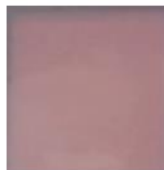
turbid

Pair 300.jpg



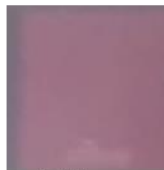
turbid

Pair 750.jpg



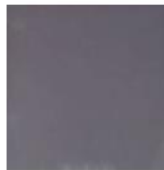
turbid

Pair 200.jpg



turbid

Pair 3670.jpg



turbid

Pair 3000.jpg



turbid

Pair 3500.jpg



turbid

Pair 20.jpg



turbid

Pair 500.jpg



turbid



clear



clear



clear



clear



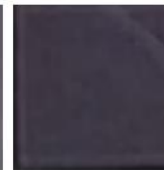
clear



clear



clear



clear



clear

Dataset pairing examples used during inspection.